

MIROSLAWIEC, Poland

WMO: 122260

Lat: 53.395N

Lon: 16.083E

Elev: 151

StdP: 99.53

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-11.6	-17.2	0.8	-13.1	-14.4	1.1	-10.6	9.8	4.0	8.8	2.5	2.0	120	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.6	19.2	27.7	18.4	25.9	17.6	20.7	27.0	19.7	25.4	18.7	23.5	3.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.7	22.6	17.8	13.0	21.7	16.9	12.3	20.6	60.2	27.3	56.8	25.6	53.8	23.4	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.6	DB	-18.0	33.1	4.1	1.9	-20.9	34.5	-23.3	35.7	-25.6	36.8	-28.5	38.2	(4)
(5)			WB	-18.2	22.7	4.1	1.0	-21.1	23.5	-23.5	24.1	-25.9	24.6	-28.8	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-2.1	-0.4	2.9	8.2	12.6	16.0	18.4	17.8	13.4	8.3	3.9	0.4	(6)	
	DBStd	8.07	5.24	4.73	4.47	4.12	4.14	3.47	3.24	3.15	3.21	3.57	3.39	4.55	(7)	
	HDD10.0	1554	374	291	222	81	19	1	0	0	6	77	184	298	(8)	
	HDD18.3	3756	633	524	479	304	183	88	41	49	153	312	434	557	(9)	
	CDD10.0	942	0	0	2	27	101	181	259	242	107	23	1	0	(10)	
	CDD18.3	102	0	0	0	6	17	42	33	4	0	0	0	0	(11)	
	CDH23.3	1089	0	0	0	9	81	212	447	298	42	0	0	0	(12)	
	CDH26.7	286	0	0	0	1	14	52	133	79	8	0	0	0	(13)	
(14) Wind	WSAvg	3.1	3.3	3.4	3.3	3.3	3.0	3.0	2.8	2.7	2.8	3.0	3.2	3.4	(14)	
(15) Precipitation	PrecAvg	676	52	38	47	38	63	71	81	77	60	48	50	52	(15)	
	PrecMax	1055	125	82	101	92	130	218	154	177	149	125	102	96	(16)	
	PrecMin	480	4	6	11	7	33	22	15	13	16	15	3	11	(17)	
	PrecStd	133	32	18	24	21	27	37	37	37	26	27	21	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	10.9	17.5	24.2	28.5	31.2	32.6	32.2	27.8	19.9	12.7	11.3	(19)	
		MCWB	8.3	8.8	10.4	14.1	17.9	20.6	19.7	20.6	19.0	14.7	10.8	9.5	(20)	
	2%	DB	7.7	9.0	14.2	21.2	25.6	27.9	29.9	28.8	24.0	17.7	10.8	9.0	(21)	
		MCWB	6.7	6.9	8.2	12.8	16.5	18.1	19.2	18.9	16.6	14.0	9.5	8.0	(22)	
	5%	DB	6.2	7.4	11.6	18.7	23.2	25.8	28.0	26.5	21.9	15.7	9.7	7.5	(23)	
		MCWB	5.3	5.7	7.5	11.4	15.5	17.1	18.6	18.1	15.5	12.8	8.7	6.5	(24)	
	10%	DB	4.7	5.8	9.4	16.3	20.8	23.5	25.8	24.5	19.6	14.0	8.7	6.0	(25)	
		MCWB	3.9	4.4	6.5	10.1	14.1	16.3	18.0	17.5	14.7	12.0	7.8	5.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.4	11.2	15.5	19.3	22.0	21.7	21.7	20.2	15.9	11.1	10.0	(27)	
		MCDB	10.1	10.4	15.4	22.8	25.1	29.0	28.0	28.3	26.8	18.7	12.1	11.0	(28)	
	2%	WB	6.8	7.4	9.6	13.6	17.6	19.6	20.6	20.4	17.6	14.4	9.7	8.2	(29)	
		MCDB	7.7	8.5	12.5	19.2	23.9	25.4	27.1	26.2	21.7	16.6	10.5	9.1	(30)	
	5%	WB	5.4	5.9	8.1	12.1	16.4	18.3	19.7	19.4	16.5	13.2	8.9	6.6	(31)	
		MCDB	6.1	7.1	10.9	17.3	21.7	23.6	25.9	24.2	20.3	15.3	9.6	7.3	(32)	
	10%	WB	3.9	4.4	6.8	10.7	15.0	17.2	18.8	18.5	15.4	12.1	8.0	5.2	(33)	
		MCDB	4.7	5.7	9.0	15.3	19.7	21.8	23.7	22.7	18.8	13.8	8.6	5.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	6.0	8.3	11.2	11.2	11.0	10.6	10.7	9.8	7.7	4.7	4.3	(35)	
		MCDBR	4.9	7.6	12.7	15.2	14.5	14.6	14.5	13.9	13.5	10.6	5.9	5.3	(36)	
	5% WB	MCWBR	4.4	5.7	7.8	8.3	7.1	6.6	5.6	6.2	6.9	6.9	4.9	4.6	(37)	
		MCWBR	4.9	6.8	10.7	13.7	12.9	12.5	12.3	11.4	11.7	9.2	5.5	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.306	0.333	0.366	0.405	0.403	0.399	0.422	0.419	0.377	0.359	0.338	0.303	(40)	
		taud	2.299	2.302	2.269	2.226	2.281	2.316	2.266	2.282	2.381	2.402	2.365	2.313	(41)	
	Ebn at Noon		659	760	809	822	846	852	822	800	793	723	617	587	(42)	
		Edh at Noon	60	82	105	124	124	121	125	117	94	75	58	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.59	1.28	2.46	4.13	5.13	5.39	5.00	4.31	3.14	1.67	0.68	0.41	(44)	
	RadStd		0.08	0.18	0.33	0.57	0.59	0.45	0.65	0.41	0.36	0.24	0.09	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-146	+63	N/A

Nomenclature: See separate page